DESIGN AND IMPLEMENTATION OF A SECURE AND ANALYTICS-DRIVEN AGRICULTURAL DATA MANAGEMENT SYSTEM USING BLOCKCHAIN TECHNOLOGY

A PROJECT REPORT

Submitted by

KASSIM ISKANDA DEISHINI
(20230404113)

in partial fulfillment for the award of the degree of

BACHELOR OF SCIENCE

in

COMPUTER SCIENCE

SCHOOL OF COMPUTING AND INFORMATION SCIENCES, DEPARTMENT OF COMPUTER SCIENCE

UNIVERSITY OF TECHNOLOGY AND APPLIED SCIENCES

JUNE, 2026

CHAPTER ONE

INTRODUCTION

1.1 Background of the study

Farming today faces a major challenge: producing enough food for a growing population while dealing with the unpredictable effects of climate change. In both developed and developing countries, unexpected weather shifts like sudden droughts or heavy rainfall make it hard to keep crop yields and supply chains stable. For local smallholder farmers, who grow a large portion of the world's food, these problems are even worse because they usually don't have access to the right tools to plan ahead. Instead, they operate in a system where markets are opaque, resources aren't always shared fairly, and powerful middlemen hold most of the control. Because of this, agriculture needs to move away from relying on guesswork and instead use data-driven systems that help farmers adapt to changing conditions.

To help solve these problems, the industry has turned to digital agriculture, which relies on tracking data about crop cycles, logistics, and the environment. Today's farms can generate a lot of data, but the main issue is no longer just gathering this information; it is figuring out how to manage and trust it. Right now, food supply chains are very fragmented. They still depend heavily on manual paperwork and isolated databases that don't communicate with each other, which makes the whole process lack transparency (Sharma, Sharma, Bhatia, & Singh, 2023). When data is locked inside disconnected systems, it becomes incredibly difficult for anyone to accurately track where a batch of crops came from, its quality, or how it was handled during transit.

The systems we currently use to manage agricultural data actually make these inefficiencies worse. Most local farming databases run on standard, centralized servers. While these are fine for basic record-keeping, they have a major structural weakness: a single point of failure. This setup leaves historical crop records, logistics data, and financial details highly vulnerable to being hacked, altered, or manipulated from the inside (Al-Absi, Al-Absi, & Lee, 2023). Furthermore, these traditional databases are mostly used just to store old information, not to make future predictions. Because the stored data can be easily changed by middlemen who might want to cheat smallholder farmers on quality or price, any analytical tools connected to these databases are also unreliable. If you cannot guarantee that the data hasn't been tampered with, any yield forecasts or automated decisions based on that data cannot be trusted.

To fix these security issues, decentralized technologies like blockchain and distributed ledgers have become a strong alternative. A blockchain acts as a secure, append-only digital record. This means that once environmental data or transaction details are saved, they are cryptographically locked and cannot be changed or deleted later (Al-Absi, Al-Absi, & Lee, 2023). Early versions of these tracking tools have already proven to be highly secure in complex industries like international shipping (Liu, Zhang, & Zhen, 2023). Building on that success, researchers are now looking at how to use these tamper-proof systems to track food and verify digital identities from the farm all the way to the consumer (Cocco, Tonelli, & Marchesi, 2021). Furthermore, recent studies show that when you combine this unalterable data collection with Big Data Analytics (BDA), you get a very powerful tool for optimizing supply chains (Chauhan, et al., 2025). By feeding secure, verified environmental data into an analytical engine, developers can build predictive models that forecast crop yields accurately, completely safe from database manipulation.

Even though blockchain and analytics offer great solutions, there is still a noticeable gap when it comes to actually using them in local farming communities. Most existing public blockchains are built for massive enterprises; they require too much computing power, have high transaction fees, and are too complex for smallholder farmers to use. On the other hand, the simpler, local platforms available to these farmers usually lack strong data security and don't have built-in forecasting tools. There simply isn't a unified system that bridges this gap by combining lightweight security (like SHA-256 hashing) with an advanced data analytics engine.

This research project addresses that exact gap by designing a secure, analytics-driven agricultural data management system. By building a secure backend ledger that inherently protects the data it holds, the system creates a trustworthy foundation for making predictions. This setup is necessary because it not only protects smallholder farmers from being exploited by providing a clear, unchangeable supply chain record, but it also gives them the reliable, data-driven insights they need to manage modern farming risks all without the heavy costs of an enterprise-level blockchain network.

1.2 Problem Statement

While basic management software is common in modern AgTech, most current platforms suffer from severe architectural and functional limitations. Existing agricultural data systems are typically designed as disjointed archiving tools. They collect scattered metrics such as environmental data, daily farm records, crop monitoring logs, and basic production information but fail to manage this data holistically. Consequently, agricultural stakeholders are forced to rely on disconnected data silos that hinder efficient on-farm decision-making and prevent true end-to-end visibility from the initial planting phases through the broader supply chain.

Compounding these operational limitations is a fundamental flaw in how production and logistics data is secured. The majority of localized agricultural databases still run on centralized relational architectures, creating a critical single point of failure. This leaves vital farm records, environmental monitoring logs, and resource allocations highly vulnerable to unauthorized retroactive edits, internal fraud, or external security breaches (Al-Absi, Al-Absi, & Lee, 2023). This lack of auditability compromises the integrity of the entire agricultural lifecycle, making it exceptionally difficult to verify historical crop conditions, validate sustainable farming practices, or ensure fair trade, ultimately leaving smallholder producers unprotected.

Furthermore, when advanced analytics and yield forecasting tools are applied to these centralized systems, their outputs become fundamentally flawed due to the vulnerability of the underlying data. Without a secure, unalterable foundation, the real agricultural data and crop monitoring profiles fed into these models can be manipulated or degraded by poor data handling. If these foundational inputs lack integrity, smallholder farmers cannot depend on the resulting production analytics and crop predictions to optimize their harvests or protect their livelihoods against climate shifts (Al Layek, Kamal, & Hasan, 2025).

Despite advancements in both database management and data science, a critical research gap remains. Existing agricultural information systems often provide either data storage capabilities or analytical services, but few integrate blockchain-based security mechanisms with predictive analytics within a unified architecture. There is a clear need for a secure, integrated platform that anchors real-time agricultural data spanning both on-farm production and supply chain logistics into an unalterable ledger before processing it. By doing so, this project ensures that all farm management decisions and predictive insights are derived from a mathematically verifiable source, directly justifying the proposed system.

1.3 Research Questions

To address the challenges identified in the problem statement, this study seeks to answer the following core research question: How can a secure, analytics-driven agricultural data management system be designed and implemented using blockchain principles to ensure data immutability, track production logistics, and provide verifiable crop yield forecasting? Specifically, the study answers the following questions:

1. How can a modular web interface be designed to effectively capture and visualize key on-farm agricultural vectors, such as weather configurations, soil quality indexes, and production parameters?
2. How can predictive statistical models be integrated into an analytical data engine to accurately calculate crop yield estimations utilizing historical agricultural datasets and modeled environmental conditions?
3. What is the most effective approach to construct an unalterable logging module that tracks the custody updates and operational transitions of produce batches across stakeholders?
4. How can SHA-256 cryptographic hashing algorithms be utilized to develop a localized backend ledger that renders the agricultural database resilient against retroactive tampering?
5. How does the integrated platform perform in terms of computational transaction latencies, block verification speeds, and prediction accuracy?

1.4 Objectives of the Study

1.4.1 General Objective

The primary objective of this project is to design and implement a secure analytics-driven agricultural data management system using blockchain technology for ensuring data integrity, traceability, and intelligent agricultural decision-making.

1.4.2 Specific Objectives

i. To design a web-based agricultural data management platform for collecting, storing, and visualizing environmental and production-related data.

ii. To develop an analytics module that applies predictive models for agricultural forecasting and decision support.
iii. To implement blockchain-based security mechanisms for ensuring data integrity, traceability, and protection against unauthorized modification of agricultural records.
iv. To evaluate the performance of the proposed system based on security effectiveness, transaction processing efficiency, and prediction accuracy.

1.5 Scope of the Study

This study has contributions to the practical, technical, and academic fields. In reality, the proposed system offers a safe and secure repository for storing, managing, and verifying agricultural data for farmers and agricultural stakeholders, and can offer predictive analytics to support data-driven decision-making. The blockchain-based architecture enhances transparency, traceability, and trust among all stakeholders, reducing the likelihood of any unauthorized changes to the information relating to agriculture. The system offers better visibility of the agricultural production and distribution process for agricultural organizations and operators of the supply chain. On a computer science basis, the study shows the design and implementation of a lightweight blockchain-based data management architecture along with analytics capabilities in a web environment. The results offer an understanding of the possibility of integrating cryptographic security mechanisms with predictive models to construct secure and intelligent agricultural information systems.

1.6 Limitations of the Study

There are a number of technical and practical constraints in this study. First, the environmental data acquisition part is modeled and not using data provided by the deployed IoT devices because of limited access to physical agricultural infrastructure. Second, system evaluation is based on historical agricultural data and scenarios of the supply chain are simulated, but not deployed in actual commercial agricultural enterprises. Third, the blockchain component used in the system is a lightweight cryptographic ledger system and does not include any distributed consensus protocols like Proof-of-Work and Proof-of-Stake that are used in large-scale blockchain networks. Fourth, the system interface is designed mainly in English which may contribute to the reduced usability of the system for users who have limited English proficiency.

Lastly, the predictive analytics models rely on the quality and representativeness of the datasets available, so they may not perform as well when extreme climatic events occur or when new weather conditions are encountered.

1.7 Organization of the Study

This thesis project report is organized into five structured chapters. Chapter One presents the introduction, gives the context of the study, identifies the core architectural problems, maps out the research objectives, and defines the technical limitations. Chapter Two presents a comprehensive Literature Review, analyzing previous peer-reviewed research regarding precision agtech, applied forecasting algorithms, and cryptographic ledger designs. Chapter Three defines the System Methodology, illustrating the data flow maps, entity-relationship models, and software choices. Chapter Four focuses on system implementation and evaluation, explaining user interfaces, core hashing algorithms, and performance profiles. Chapter Five provides a complete summary of findings, draws conclusions, and outlines explicit recommendations for future technical updates.

1.8 Chapter Summary

This chapter established the groundwork for the project by introducing a secure, blockchain oriented agricultural data management framework. It reviewed the operational flaws found in centralized AgTech databases, focusing on data vulnerability and the lack of trusted predictive analytics. The chapter clearly detailed the specific programming objectives driving the creation of the prototype, illustrated the practical value of the platform for farmers and software engineers, defined the explicit scope boundaries of the study, and provided a structural overview for the remaining chapters of this thesis.

References

Al Layek, M. A., Kamal, M. S., & Hasan, M. F. (2025). A Secured Triad of IoT, Machine Learning, and Blockchain for Crop Forecasting in Agriculture. *arXiv preprint arXiv:2505.01196*.

Al-Absi, A. A., Al-Absi, M. A., & Lee, J. (2023). Blockchain-Based Smart Farm Security Framework for the Internet of Things. *Sensors, 23*(18), 7992. Retrieved from https://doi.org/10.3390/s23187992

Chauhan, V., Jain, N., Waghmare, M., Parmar, A., Nandeha, N., Trivedi, A., . . . Rahman, S. W. (2025). Blockchain and Big Data Analytics in Agriculture: A Review of Digital Innovations. *Journal of Experimental Agriculture International, 47*(11), 58-68. Retrieved from https://doi.org/10.9734/jeai/2025/v47i113849

Cocco, L., Tonelli, R., & Marchesi, M. (2021). Blockchain and Self Sovereign Identity to Support Quality in the Food Supply Chain. *Future Internet, 13*(12), 301. Retrieved from https://doi.org/10.3390/fi1312

Liu, J., Zhang, H., & Zhen, L. (2023). Blockchain technology in maritime supply chains: Applications, architecture and challenges. *International Journal of Production Research, 61*(11), 3547-3563. Retrieved from https://doi.org/10.1080/00207543.2021.1930239

Sharma, A., Sharma, A., Bhatia, T., & Singh, R. K. (2023). Blockchain enabled food supply chain management: A systematic literature review and bibliometric analysis. *Operations Management Research, 16*(3), 1594-1618. doi:10.1007/s12063-02300361-9